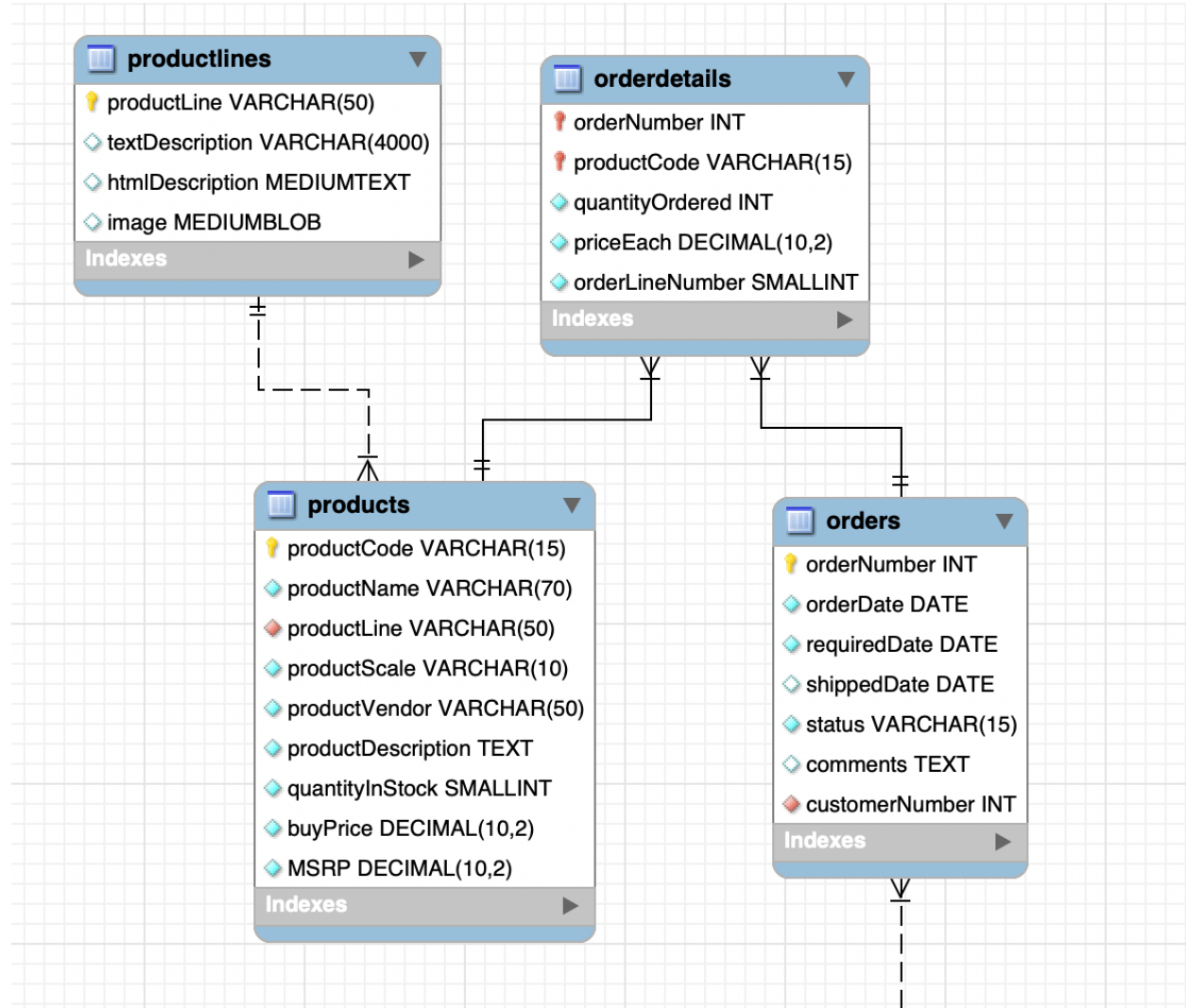


Assignment 3
ISAN 5355 with Dr. Zhao
John Courtright

ERR Diagram (for my own sake):



Question 1) Display orderNumber, quantityOrdered, priceEach, productName for all orders

```
SELECT od.orderNumber, od.quantityOrdered, od.priceEach, p.productName
FROM orderdetails as od
JOIN products as p USING(productCode);
```

```
10  /* 1) Display orderNumber, quantityOrdered, priceEach, productName
11      for all orders */
12  •  SELECT od.orderNumber, od.quantityOrdered, od.priceEach, p.productName
13      FROM orderdetails as od
14      JOIN products as p USING(productCode);
```

100% 70:8

Result Grid Filter Rows: Search Export: Fetch rows:

orderNumber	quantityOrder...	priceEach	productName
10107	30	81.35	1969 Harley Davidson Ultimate Chopper
10121	34	86.13	1969 Harley Davidson Ultimate Chopper
10134	41	90.92	1969 Harley Davidson Ultimate Chopper
10145	45	76.56	1969 Harley Davidson Ultimate Chopper
10159	49	81.35	1969 Harley Davidson Ultimate Chopper
10168	36	94.74	1969 Harley Davidson Ultimate Chopper
10180	20	76.56	1969 Harley Davidson Ultimate Chopper

Result 1 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	14:08:36	SELECT od....	2996 row(s) returned	0.037 sec / 0.0046 sec

Question 2) Modify Q1) to use inline view to display the orderNumber, cost per each order (using an alias 'CostPerOrder'), and the productName

```
SELECT
    sub.orderNumber, sub.CostPerOrder, p.productName
FROM (
    SELECT
        orderNumber,
        productCode,
        (quantityOrdered * priceEach) AS CostPerOrder
    FROM orderdetails
) AS sub
JOIN products AS p USING (productCode);
```

16 /* 2) Modify the query in question 1 to use inline view to display
17 the orderNumber, cost per each order (using an alias 'CostPerOrder'),
18 and the productName */
19 • **SELECT**
20 sub.orderNumber, sub.CostPerOrder, p.productName
21 **FROM** (
22 **SELECT**
23 orderNumber,
24 productCode,
25 (quantityOrdered * priceEach) **AS** CostPerOrder
26 **FROM** orderdetails
27) **AS** sub
28 **JOIN** products **AS** p **USING** (productCode);
29

100% 39:14

Result Grid Filter Rows: Search Export: Fetch rows:

orderNumber	CostPerOrder	productName
10134	3727.72	1969 Harley Davidson Ultimate Chopper
10145	3445.20	1969 Harley Davidson Ultimate Chopper
10159	3986.15	1969 Harley Davidson Ultimate Chopper
10168	3410.64	1969 Harley Davidson Ultimate Chopper
10180	2220.24	1969 Harley Davidson Ultimate Chopper
10188	4593.60	1969 Harley Davidson Ultimate Chopper

Result 3 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	14:26:00	SELECT...	2996 row(s) returned	0.015 sec / 0.0034 sec

Question 3) Display orderNumber, productName, orderDate for all orders placed after May 30th, 2005 (orderDate > 2005-05-30)

```
SELECT od.orderNumber, o.orderDate, p.productName
FROM orderdetails as od
INNER JOIN orders as o ON od.orderNumber = o.orderNumber
INNER JOIN products as p ON od.productCode = p.productCode
WHERE o.orderDate > '2005-05-30';
```

```
30  /* 3) Display orderNumber, productName, orderDate for all orders
31      placed after May 30th, 2005 (orderDate > 2005-05-30) */
32  •  SELECT od.orderNumber, o.orderDate, p.productName
33      FROM orderdetails as od
34      INNER JOIN orders as o ON od.orderNumber = o.orderNumber
35      INNER JOIN products as p ON od.productCode = p.productCode
36      WHERE o.orderDate > '2005-05-30';
37
```

100% 40:28

Result Grid Filter Rows: Search Export:

orderNumber	orderDate	productName
10424	2005-05-31	1952 Alpine Renault 1300
10424	2005-05-31	1958 Setra Bus
10424	2005-05-31	1940 Ford Pickup Truck
10424	2005-05-31	1939 Cadillac Limousine
10424	2005-05-31	1996 Peterbilt 379 Stake Bed with Outrigger
10424	2005-05-31	1982 Camaro Z28
10425	2005-05-31	1999 Lancia Delta 16V

Result 8 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 4	15:13:14	SELECT od...	19 row(s) returned	0.019 sec / 0.00049...
✓ 5	15:13:26	SELECT od...	19 row(s) returned	0.0012 sec / 0.00001...
✓ 6	15:13:53	SELECT od....	19 row(s) returned	0.0028 sec / 0.00002...

Question 4) Modify the query above to use a subquery to obtain the same output only with the columns orderNumber, productName

```
SELECT od.orderNumber, p.productName
FROM orderdetails AS od
JOIN products AS p USING (productCode)
WHERE od.orderNumber IN (
    SELECT orderNumber
    FROM orders
    WHERE orderDate > '2005-05-30'
)
```

```
38  /* 4) Modify the query above to use a subquery to obtain the
39      same output only with the columns orderNumber, productName */
40  •  SELECT od.orderNumber, p.productName
41      FROM orderdetails AS od
42      JOIN products AS p USING (productCode)
43      WHERE od.orderNumber IN (
44          SELECT orderNumber
45          FROM orders
46          WHERE orderDate > '2005-05-30'
47      )
```

100%

1:37

Result Grid

Filter Rows:

Export:

Result Grid

Form Editor

orderNumber	productName
10424	1952 Alpine Renault 1300
10424	1958 Setra Bus
10424	1940 Ford Pickup Truck
10424	1939 Cadillac Limousine
10424	1996 Peterbilt 379 Stake Bed with Outrigger
10424	1982 Camaro Z28
10425	1999 Lexus A Delta 10V

Result 7

Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
3	15:11:14	SELECT od...	Error Code: 1241. Operand should contain 1 column(s)	0.033 sec
4	15:13:14	SELECT od...	19 row(s) returned	0.019 sec / 0.00049...
5	15:13:26	SELECT od...	19 row(s) returned	0.0012 sec / 0.00001...

Question 5) Select the productName and compute the corresponding TotalPerProduct (i.e., sum product of quantityOrdered and priceEach), grouping the outcome by the productName, for all the totals greater than 180000 using a join statement.

```
SELECT p.productName,
       SUM(od.quantityOrdered * od.priceEach) as TotalPerProduct
FROM products as p
JOIN orderdetails as od USING (productCode)
GROUP BY p.productName
HAVING SUM(od.quantityOrdered * od.priceEach) > 180000;
```

```

49  /* 5) Select the productName and compute the corresponding
50      TotalPerProduct (i.e., sum product of quantityOrdered and
51      priceEach), grouping the outcome by the productName, for
52      all the totals greater than 180,000 */
53  •  SELECT p.productName,
54         SUM(od.quantityOrdered * od.priceEach) as TotalPerProduct
55      FROM products as p
56      JOIN orderdetails as od USING (productCode)
57      GROUP BY p.productName
58      HAVING SUM(od.quantityOrdered * od.priceEach) > 180000;

```

100% 35:46

Result Grid

Filter Rows: Search

Export:

productName	TotalPerProdu...
1952 Alpine Renault 1300	190017.96
2001 Ferrari Enzo	190755.86
1992 Ferrari 360 Spider red	276839.98

Result 9 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	15:39:40	SELECT p.p...	3 row(s) returned	0.041 sec / 0.00036...

Question 6) Update the orderDate for the order with orderNumber as 10100 to be 3 days later than the current value.

UPDATE orders

SET orderDate = DATE_ADD(orderDate, INTERVAL 3 DAY)

WHERE orderNumber = 10100;

```
60  /* 6) Update the orderDate for order 10100 to be
61      3 days later than the current value. */
62  •  UPDATE orders
63      SET orderDate = DATE_ADD(orderDate, INTERVAL 3 DAY)
64      WHERE orderNumber = 10100;
65      -- View Row(s)
66  •  SELECT orderNumber, orderDate
67      FROM orders
68      WHERE orderNumber = 10100;
```

100% 27:68

Result Grid Filter Rows: Search Edit: Export/Import:

orderNumber	orderDate
10100	2003-01-09
NULL	NULL

orders 10 Apply

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	15:47:31	UPDATE or...	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.051 sec
✓ 2	15:47:31	SELECT ord...	1 row(s) returned	0.0024 sec / 0.00062...

Question 7) Find all the orders that were placed in July

```
SELECT *  
FROM orders  
WHERE MONTH(orderDate) = 7;
```

```
70      /* 7) Find all the orders that were placed in July */  
71 •    SELECT *  
72      FROM orders  
73      WHERE MONTH(orderDate) = 7;
```

100% 27:68

Result Grid

Filter Rows: Search

Edit: Export/Import:

orderNumber	orderDate	requiredDate	shippedDate	status	comments
10134	2003-07-01	2003-07-10	2003-07-05	Shipped	NULL
10135	2003-07-02	2003-07-12	2003-07-03	Shipped	NULL
10136	2003-07-04	2003-07-14	2003-07-06	Shipped	Customer is interested in buying more Ferrari
10137	2003-07-10	2003-07-20	2003-07-14	Shipped	NULL
10138	2003-07-07	2003-07-16	2003-07-13	Shipped	NULL
10139	2003-07-16	2003-07-23	2003-07-21	Shipped	NULL
10140	2003-07-04	2003-07-08	2003-07-08	Shipped	NULL

orders 11

Apply

Result Grid

Form Editor

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	15:50:22	SELECT * F...	18 row(s) returned	0.025 sec / 0.00016 s...

Question 8) Find all the orders that were placed on the 26th day of any month in 2003.

```
SELECT *  
FROM orders  
WHERE DAYOFMONTH(orderDate) = 26  
      AND YEAR(orderDate) = 2003;
```

75 /* 8) Find all the orders that were placed on the 26th day
76 of any month in 2003 */
77 • SELECT *
78 FROM orders
79 WHERE DAYOFMONTH(orderDate) = 26
80 AND YEAR(orderDate) = 2003;

100% 34:70

Result Grid Filter Rows: Search Edit: Export/Import:

orderNumber	orderDate	requiredDate	shippedDate	status	comments
10113	2003-03-26	2003-04-02	2003-03-27	Shipped	NULL
10196	2003-11-26	2003-12-03	2003-12-01	Shipped	NULL
10197	2003-11-26	2003-12-02	2003-12-01	Shipped	Customer inquired about remote controlled mo
NULL	NULL	NULL	NULL	NULL	NULL

orders 13 Apply

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	15:54:52	SELECT * F...	3 row(s) returned	0.0012 sec / 0.00001...

Question 9) Find the productName and TotalPerProduct (i.e., sum product of quantityOrdered and priceEach) for all products whose names start with 1957, followed by a space, and then a letter from A to G.

```
SELECT p.productName,
       SUM(od.quantityOrdered * od.priceEach) AS TotalPerProduct
FROM products AS p
JOIN orderdetails AS od USING (productCode)
WHERE p.productName REGEXP '^1957 [A-G]'
GROUP BY p.productName;
```

```

82  /* 9) Find the productName and TotalPerProduct for all products
83      whose names start with 1957, followed by a space, and
84      then a letter from A to G */
85  •  SELECT p.productName,
86         SUM(od.quantityOrdered * od.priceEach) AS TotalPerProduct
87  FROM products AS p
88  JOIN orderdetails AS od USING (productCode)
89  WHERE p.productName REGEXP '^1957 [A-G]'
90  GROUP BY p.productName;

```

100%
29:80

Result Grid
Filter Rows:
Export:

productName	TotalPerProdu...
1957 Chevy Pickup	109946.21
1957 Corvette Convertible	130749.31
1957 Ford Thunderbird	50101.57

Result 14
 Read Only

Action Output
⌵

	Time	Action	Response	Duration / Fetch Time
✓ 1	16:03:08	SELECT p.p...	3 row(s) returned	0.048 sec / 0.00032...

Question 10) Convert all orderNumber from integers to strings with a length of 10 characters

-- Create new columns in orders and orderdetails

ALTER TABLE orders

ADD orderNumberChar Char(10);

ALTER TABLE orderdetails

ADD orderNumberChar Char(10);

Add New Rows

```
93  /* 10) Convert all orderNumber from integers to
94      strings with a length of 10 characters */
95      -- Create new columns in orders and orderdetails
96  • ALTER TABLE orders
97      ADD orderNumberChar Char(10);
98  • ALTER TABLE orderdetails
99      ADD orderNumberChar Char(10);
100  -- Verify
101  • SELECT orderNumberChar
102      FROM orders;
103  • SELECT orderNumberChar
104      FROM orderdetails;
105
```

100% 29:92

Result Grid Filter Rows: Search Export: Fetch rows:

orderNumberChar...

orders 20 orderdetails 21 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
2	18:51:03	ALTER TABLE orderdetails ADD orderNumberChar Char(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.00064 sec
3	18:51:07	SELECT orderNumberChar FROM orders	326 row(s) returned	0.0040 sec / 0.00079...
4	18:51:07	SELECT orderNumberChar FROM orderdetails	2996 row(s) returned	0.0017 sec / 0.00035...

Insert new, converted orderNumbers with LPAD to add 0's to pad out the value

UPDATE orderdetails

SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);

UPDATE orders

SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);

```
105 -- Add new values to columns
106 • UPDATE orderdetails
107 SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);
108 • UPDATE orders
109 SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);
110 -- Print both columns to verify
111 • SELECT ordernumber, orderNumberChar
112 FROM orders;
113 • SELECT ordernumber, orderNumberChar
114 FROM orderdetails;
```

100% 23:103

Result Grid Filter Rows: Search Export: Fetch rows: Read Only

ordernumb...	orderNumberCh...
10100	0000010100
10100	0000010100
10100	0000010100
10100	0000010100
10101	0000010101
10101	0000010101

orders 22 orderdetails 23

Action Output

	Time	Action	Response	Duration / Fetch Time
6	18:53:18	UPDATE orders SET orderNumberChar = LPAD(CONVERT(ordernumber, Char(10)), 10, 0)	326 row(s) affected rows matched: 326 Changed: 3...	0.0009 / sec
7	18:53:39	SELECT ordernumber, orderNumberChar FROM orders	326 row(s) returned	0.0033 sec / 0.00079...
8	18:53:39	SELECT ordernumber, orderNumberChar FROM orderdetails	2996 row(s) returned	0.0023 sec / 0.00071...

Courtright_Assign3* Class7_Demo classicmodels.orderdetails classicmodels.orders

Info Columns Indexes Triggers Foreign keys Partitions Grants DDL

Column	Type	Default Value	Nullable	Character Set	Collation	Privileges	Extra
orderNumber	int		NO			select,insert,update,references	
productCode	varchar(15)		NO	utf8mb4	utf8mb4_090...	select,insert,update,references	
quantityOrdered	int		NO			select,insert,update,references	
priceEach	decimal(10,2)		NO			select,insert,update,references	
orderLineNumber	smallint		NO			select,insert,update,references	
orderNumberChar	char(10)		YES	utf8mb4	utf8mb4_090...	select,insert,update,references	

Courtright_Assign3* Class7_Demo classicmodels.orderdetails classicmodels.orders

Info Columns Indexes Triggers Foreign keys Partitions Grants DDL

Column	Type	Default Value	Nullable	Character Set	Collation	Privileges	Extra
orderNumber	int		NO			select,insert,update,references	
orderDate	date		NO			select,insert,update,references	
requiredDate	date		NO			select,insert,update,references	
shippedDate	date		YES			select,insert,update,references	
status	varchar(15)		NO	utf8mb4	utf8mb4_090...	select,insert,update,references	
comments	text		YES	utf8mb4	utf8mb4_090...	select,insert,update,references	
customerNumber	int		NO			select,insert,update,references	
orderNumberChar	char(10)		YES	utf8mb4	utf8mb4_090...	select,insert,update,references	

Full SQL Script:

```
/*
    ISAN 5355: Database Management Systems
    Assignment 3: Due 10/29
    Author: John Courtright
    Instructor: Dr. Zhao
*/

-- SET sql_mode=(SELECT REPLACE(@@sql_mode,'ONLY_FULL_GROUP_BY',''));

/* 1) Display orderNumber, quantityOrdered, priceEach, productName
    for all orders */
SELECT od.orderNumber, od.quantityOrdered, od.priceEach, p.productName
FROM orderdetails as od
JOIN products as p USING(productCode);

/* 2) Modify the query in question 1 to use inline view to display
    the orderNumber, cost per each order (using an alias 'CostPerOrder'),
    and the productName */
SELECT
    sub.orderNumber, sub.CostPerOrder, p.productName
FROM (
    SELECT
        orderNumber,
        productCode,
        (quantityOrdered * priceEach) AS CostPerOrder
    FROM orderdetails
) AS sub
JOIN products AS p USING (productCode);

/* 3) Display orderNumber, productName, orderDate for all orders
    placed after May 30th, 2005 (orderDate > 2005-05-30) */
SELECT od.orderNumber, o.orderDate, p.productName
FROM orderdetails as od
INNER JOIN orders as o ON od.orderNumber = o.orderNumber
INNER JOIN products as p ON od.productCode = p.productCode
WHERE o.orderDate > '2005-05-30';

/* 4) Modify the query above to use a subquery to obtain the
    same output only with the columns orderNumber, productName */
SELECT od.orderNumber, p.productName
FROM orderdetails AS od
JOIN products AS p USING (productCode)
```

```
WHERE od.orderNumber IN (
    SELECT orderNumber
    FROM orders
    WHERE orderDate > '2005-05-30'
);
```

```
/* 5) Select the productName and compute the corresponding
    TotalPerProduct (i.e., sum product of quantityOrdered and
    priceEach), grouping the outcome by the productName, for
    all the totals greater than 180,000 */
SELECT p.productName,
    SUM(od.quantityOrdered * od.priceEach) as TotalPerProduct
FROM products as p
JOIN orderdetails as od USING (productCode)
GROUP BY p.productName
HAVING SUM(od.quantityOrdered * od.priceEach) > 180000;
```

```
/* 6) Update the orderDate for order 10100 to be
    3 days later than the current value. */
UPDATE orders
SET orderDate = DATE_ADD(orderDate, INTERVAL 3 DAY)
WHERE orderNumber = 10100;
-- View Row(s)
SELECT orderNumber, orderDate
FROM orders
WHERE orderNumber = 10100;
```

```
/* 7) Find all the orders that were placed in July */
SELECT *
FROM orders
WHERE MONTH(orderDate) = 7;
```

```
/* 8) Find all the orders that were placed on the 26th day
    of any month in 2003 */
SELECT *
FROM orders
WHERE DAYOFMONTH(orderDate) = 26
    AND YEAR(orderDate) = 2003;
```

```
/* 9) Find the productName and TotalPerProduct for all products
    whose names start with 1957, followed by a space, and
    then a letter from A to G */
SELECT p.productName,
    SUM(od.quantityOrdered * od.priceEach) AS TotalPerProduct
```

```
FROM products AS p
JOIN orderdetails AS od USING (productCode)
WHERE p.productName REGEXP '^1957 [A-G]'
GROUP BY p.productName;

-- SET SQL_SAFE_UPDATES = 0;
/* 10) Convert all orderNumber from integers to
   strings with a length of 10 characters */
-- Create new columns in orders and orderdetails
ALTER TABLE orders
ADD orderNumberChar Char(10);
ALTER TABLE orderdetails
ADD orderNumberChar Char(10);
-- Verify
SELECT orderNumberChar
FROM orders;
SELECT orderNumberChar
FROM orderdetails;
-- Add new values to columns
UPDATE orderdetails
SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);
UPDATE orders
SET orderNumberChar = LPAD(CONVERT(orderNumber, Char(10)), 10, 0);
-- Print both columns to verify
SELECT ordernumber, orderNumberChar
FROM orders;
SELECT ordernumber, orderNumberChar
FROM orderdetails;
```